

Advanced (Habanero)

Q1. Solve the system: $x + y = 3$, $x^2 + y^2 = 9$

- (A) $(x, y) = (1, 2)$
(B) $(x, y) = (2, 1)$
(C) $x = 1, y = 2$ or $x = 2, y = 1$
(D) $x = -1, y = 4$ or $x = 4, y = -1$
(E) No solution

Q2. Find the intersection point: $y = 2x - 3$, $y = x^2 - 4$

- (A) $(-1, -5)$
(B) $(2, 1)$
(C) $(-2, -1)$
(D) $(1, -1)$
(E) $(0, -3)$

Q3. Solve the system: $y = x + 1$, $y = -x^2 + 2$

- (A) $x = -1, y = 0$
(B) $x = 1, y = 2$
(C) $x = 0, y = 1$
(D) $x = -2, y = -1$
(E) $x = 2, y = 3$

Q4. Find the value of x for which the system has a unique solution: $y = x^2 + 2x + 1$, $y = -x + 3$

- (A) $x = -1$
(B) $x = 1$
(C) $x = 2$
(D) $x = -2$
(E) $x = 0$

Q5. Solve the system: $x^2 - y^2 = 4$, $x - y = 2$

- (A) $x = 2, y = 0$
(B) $x = 0, y = -2$
(C) $x = 3, y = 1$
(D) $x = 1, y = -1$
(E) $x = -1, y = 1$

Q6. Find the intersection points: $y = 2x^2 - 3x + 1$, $y = x + 2$

- (A) $(-1, 1)$, $(1, 3)$
(B) $(-1, 1)$, $(2, 4)$
(C) $(0, 1)$, $(2, 5)$
(D) $(1, 3)$, $(2, 4)$
(E) $(0, 2)$, $(1, 3)$

Q7. Solve the system: $x + y = 1$, $x^2 - y^2 = -3$

- (A) $x = -1, y = 2$
(B) $x = 2, y = -1$
(C) $x = 1, y = 0$
(D) $x = 0, y = 1$ or $x = -1, y = 2$
(E) $x = 1, y = 0$ or $x = -2, y = 3$

Q8. Find the y -coordinate of the intersection point: $y = x^2 - 2$, $y = 2x - 1$

- (A) $y = -1$
(B) $y = 0$
(C) $y = 1$
(D) $y = 2$
(E) $y = 3$

Open Ended Questions (FRQ)**Q9.** Solve the system of equations: $y = x^2 + x - 2$, $y = -2x + 3$ **Q10.** Find the intersection points of the system: $x^2 + y^2 = 4$, $y = x - 1$ **Chef's Tips**

- Be sure to check for extraneous solutions
- Consider using substitution or elimination methods
- Graphical analysis may be useful for visualizing the solutions